Solving Initial Value Ordinary Differential Equations

Order and initial conditions

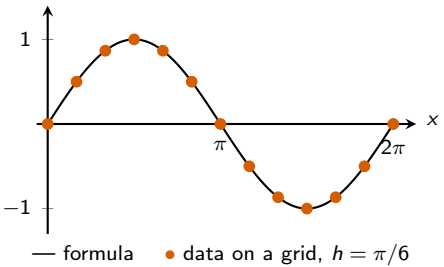
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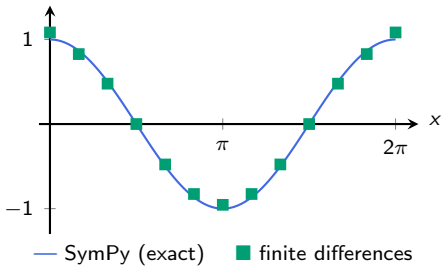
October 23, 2026

So far: derivatives, two ways

given: $f(x) = \sin x$



derivative: $f'(x)$



Next: same two routes, but now the **unknown** is the function $y(t)$ itself.

Ordinary differential equations: order and initial conditions

An **ordinary differential equation** (ODE) links an unknown $y(t)$ to its derivatives. **Order** = the highest derivative that appears.

First order: one initial condition

$$y'(t) = F\big(t, y\big), \quad y(t_0) = y_0$$

e.g. $y' + \frac{1}{2}y = \sin t, \quad y(0) = 1$

General solution has **one** constant C_1.

Second order: two initial conditions

$$y''(t) = F\big(t, y, y'\big), \quad y(t_0) = y_0,\ y'(t_0) = y_0'$$

e.g. $y'' + 2y' + 5y = 0, \quad y(0) = 1,\ y'(0) = 0$

Two constants C_1, C_2: start position *and* velocity.

- The equation alone gives a **family** of solutions; the initial conditions pick **one**.

order n $\Rightarrow$ n constants $\Rightarrow$ n initial conditions